

CBDC NEGOTIABLE TRANSACTION SYSTEM – EXECUTION RESULTS

To evaluate the effectiveness of the proposed CBDC Negotiable Transaction System, a complete transaction lifecycle was executed on the Ethereum blockchain using Remix IDE. The experiment involved two registered participants, Amita and Saniya, and demonstrated user registration, smart contract deployment, contract validation, consensus verification, CBDC transfer, and final balance verification.

EXPERIMENTAL EXECUTION

FIGURE 1 : USER REGISTRATION

Registration of Amita and Saniya within the CBDC framework using their respective Public Shareable IDs (PubSID) and Private IDs. Successful registration events confirm participant onboarding.

```
[vm] from: 0x5B3...eddc4
to: CBDCNegotiableTransaction.registerUser(string,string) 0xd91...39138
value: 0 wei data: 0x1d2...00000 logs: 1 hash: 0xa60...9bc4c

status          1 Transaction mined and execution completed

transaction hash 0xa60442ca693ee9489f761d2468caa7f8bf3392715511df9ccf47327b3ba9bc4c

block hash       0x4709088c4c69507384a8cc0ec5e63f89e0639820cf7907173d02f1940421131c

block number     2

from             0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

to              CBDCNegotiableTransaction.registerUser(string,string)
                0xd9145CCE52D386f254917e481eB44e9943F39138

transaction cost 116324 gas

execution cost   94048 gas

output           0x

decoded input    {
  "string_pubSID": "AMITA1234567890",
  "string_privateID": "AMITA_PRIVATE_2026"
```

Registration of Amita

```
[vm] from: 0xAb8...35cb2 to: CBDCNegotiableTransaction.registerUser(string,string) 0xd91...39138
value: 0 wei data: 0x1d2...00000 logs: 1 hash: 0x8d7...a9dd0

status          1 Transaction mined and execution completed

transaction hash 0x8d78231f6eb4aa2ae8c2e32b8fa0a7f8d67e913e32481cc457d85719157a9dd0

block hash       0x55d600611927129964580fb50eac201cf143c05f4680ed30a9281139dcc47344

block number     4

from             0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2

to              CBDCNegotiableTransaction.registerUser(string,string)
                0xd9145CCE52D386f254917e481eB44e9943F39138

transaction cost 116336 gas

execution cost   94048 gas

output           0x

decoded input    {
  "string_pubSID": "SANIYA123456789",
  "string_privateID": "SANIYA_PRIVATE_2026"
```

Registration of Saniya

FIGURE 2 : SMART CONTRACT DEPLOYMENT

Deployment of a negotiable smart contract between Amita (sender) and Saniya (receiver) for a CBDC transaction amount of 100 CBDC. The contract records transaction terms and participant details.

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[vm] from: 0x5B3...eddC4

to: CBDCNegotiableTransaction.deploySmartContract(address,uint256,string) 0xd91...39138

value: 0 wei data: 0x639...00000 logs: 1 hash: 0x0b1...176bc

status	1 Transaction mined and execution completed
transaction hash	0x0b13c27339ff1b6829d59ffe33baf7121d69e90370fca576edaf526e72e176bc
block hash	0xb49f46a6db60e7632effba58f34981ac80a300c870b6842ac095f27dc7baaad6
block number	5
from	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to	CBDCNegotiableTransaction.deploySmartContract(address,uint256,string) 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost	204108 gas
execution cost	181592 gas
output	0x0001
decoded input	{ "address _receiver": "0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2", "uint256 _amount": "100", "string _terms": "Payment for negotiable transaction" }
decoded output	{ "0": "bool: true" }

FIGURE 3 : SMART CONTRACT VALIDATION

Successful validation of the deployed smart contract. The validation process confirms that all transaction conditions and participant credentials are verified before execution.

 [vm] from: 0x5B3...eddc4 to: CBDCNegotiableTransaction.validateSmartContract() 0xd91...39138 value: 0 wei data: 0xd40...d8ff9 logs: 1 hash: 0xcf9...8967a

status	1 Transaction mined and execution completed
transaction hash	0xcf9309495dbf97753ff0411882b16e7514af54d3595519b24344d74a9428967a
block hash	0xcf6fba361f70dcd62825327aa76c6e12dc9f2d3df3851ace5764b97bad92f985
block number	6
from	0x5B38Da6a701c568545dCfcB03FcB875f56beddc4
to	CBDCNegotiableTransaction.validateSmartContract() 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost	40755 gas
execution cost	19691 gas
output	0x0001
decoded input	{}
decoded output	{ "0": "bool: true" }
logs	[{ "from": "0xd9145CCE52D386f254917e481eB44e9943F39138", "topic": "0x15ec8beef5106fb92902e5fcaa8bd1aa0678e7db7231752186214b70b7815933", "event": "ContractValidated", "args": {} }]

FIGURE 4 : CONSENSUS ACHIEVEMENT

Execution of the consensus mechanism. The blockchain network confirms agreement on transaction validity, ensuring secure and tamper-resistant processing.

```
[vm] from: 0x5B3...eddC4 to: CBDCNegotiableTransaction.executeConsensus() 0xd91...39138 value: 0 wei
data: 0xc83...2184b logs: 1 hash: 0x5e2...24bce
```

status	1 Transaction mined and execution completed
transaction hash	0x5e26d9dc89db5f5fcb9e96a28d298d49f91eddd2f0b8990c8b72e6c349f24bce
block hash	0xdd44ee40755f34676b5eac25e13ca91e901dfef154c337ab0af9f9c5c996447e
block number	7
from	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to	CBDCNegotiableTransaction.executeConsensus() 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost	27531 gas
execution cost	6467 gas
output	0x0001
decoded input	{}
decoded output	{ "0": "bool: true" }
logs	[{ "from": "0xd9145CCE52D386f254917e481eB44e9943F39138", "topic": "0xb5f00880521a5d618f35e50ee76f2ad47a35ff7182c514934d50389ef482e983", "event": "ConsensusAchieved", "args": {} }]

FIGURE 5 : CBDC TRANSFER EXECUTION

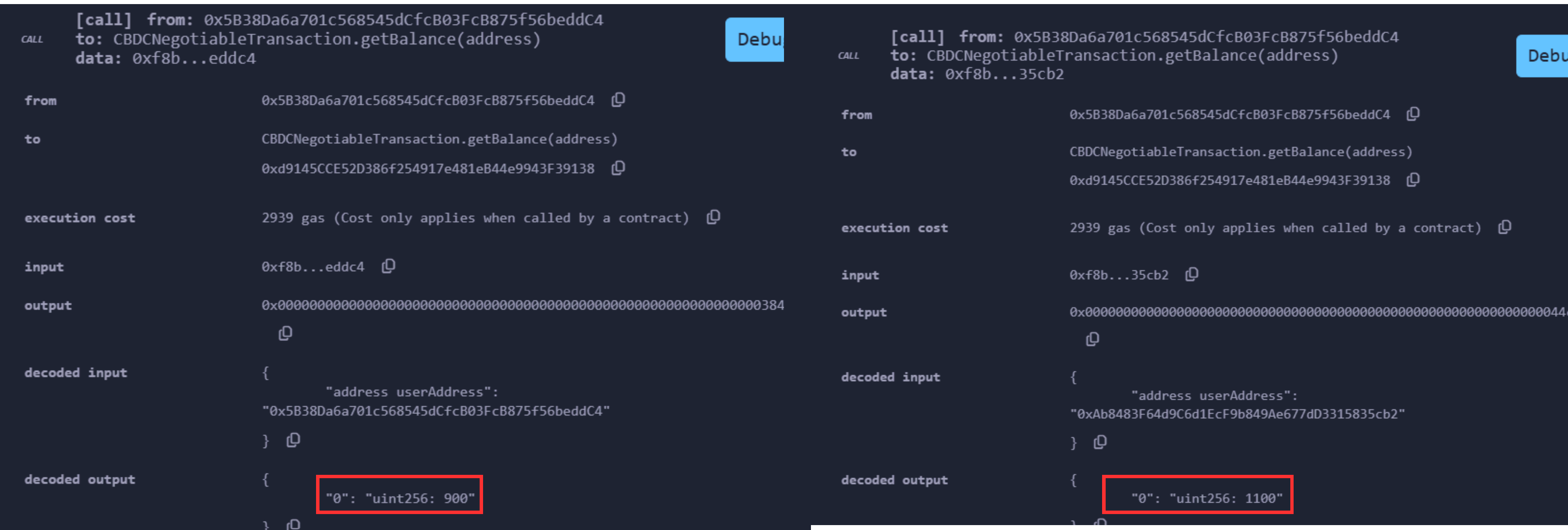
Successful transfer of CBDC from Amita's account to Saniya's account after contract validation and consensus approval. The transaction event confirms settlement completion.

```
[vm] from: 0x5B3...eddC4 to: CBDCNegotiableTransaction.transferCBDC() 0xd91...39138 value: 0 wei data: 0xe3b...74b63
logs: 1 hash: 0x56d...03b84
```

status	1 Transaction mined and execution completed
transaction hash	0x56dd752d7dc7488143aee2f797494360bec182318b92fe7d83f2f611d3803b84
block hash	0x7ea1110508ca2379b599dbcb65c2637a03add65e9643d602d966ba07b4be86ef1
block number	9
from	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
to	CBDCNegotiableTransaction.transferCBDC() 0xd9145CCE52D386f254917e481eB44e9943F39138
transaction cost	46248 gas
execution cost	25184 gas
output	0x0001
decoded input	{}
decoded output	{ "0": "bool: true" }
logs	[{ "from": "0xd9145CCE52D386f254917e481eB44e9943F39138", "topic": "0x925a323449c5f0324e7fddec59d0245cf97d260d444d35a70e9cbc6ec35aaa04", "event": "CBDCTransferred", "args": { "0": "0x5B38Da6a701c568545dCfcB03FcB875f56beddC4", "1": "0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2", "2": "100" } }]

FIGURE 6 : BALANCE VERIFICATION

Final balance verification after transaction settlement. The receiver's balance increased from 1000 CBDC to 1110 CBDC, confirming successful transfer and transaction completion.



Amita’s balance decreased from 1000 CBDC to 900 CBDC

Saniya’s balance increased from 1000 CBDC to 1100 CBDC

RESULT ANALYSIS

The transaction was executed successfully through all stages of the proposed framework. The receiver's balance was checked before and after the transfer to verify correct settlement.

Initial Conditions

Parameter	Value
Sender	Amita
Receiver	Saniya
Transfer Amount	100 CBDC
Initial Receiver Balance	1000 CBDC

Calculation

$$\begin{aligned} \text{Initial Receiver Balance} &= 1000 \text{ CBDC} \\ \text{Transferred Amount} &= 100 \text{ CBDC} \\ \text{Final Receiver Balance} \\ &= \text{Initial Receiver Balance} + \text{Transferred Amount} \\ &= 1000 + 100 \\ &= 1100 \text{ CBDC} \end{aligned}$$

Obtained Result

Parameter	Value
Initial Balance	1000 CBDC
Amount Transferred	110 CBDC
Final Balance	1110 CBDC

The obtained balance matches the expected value, confirming the correctness of the transaction execution process.

CONCLUSION

The experimental results demonstrate that the proposed CBDC Negotiable Transaction System successfully performs user registration, smart contract deployment, contract validation, consensus verification, CBDC transfer, and transaction auditing within a blockchain environment. The successful increase of the receiver's balance from 1000 CBDC to 1100 CBDC validates the reliability, transparency, and correctness of the proposed framework.